

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Under a rigid-strain upper-bound protocol (PBE+D3) with fixed fractional coordinates, all reported coupling magnitudes bound—rather than predict—equilibrium total-energy cross terms until ionic-relaxation benchmarks complete (Table II). Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects are not necessarily additive. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with the synergy order parameter $\mathcal{S}$ (meV per atom) [Eq. (1)]. On tetramers ($\sim 5\%$ nominal doping) and fifteen periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$, B/N/P @ $+3\%$, one substituent per C₆₀, ~ 1.67 at.%), the linear strain coefficient α is opposite for N (-303 meV/%) and B ($+57$ meV/%), and the protocol yields $|\mathcal{S}|$ values up to 31.9 meV/atom for P at $n=1$ and $+3\%$ strain, with P showing the largest $|\mathcal{S}|$ despite the smallest $|\alpha|$. Sequential high-throughput screens can introduce systematic errors in the estimation of *relative* stability margins when local dopant distortion couples nonlinearly to external strain. Marcus reorganization energies and nearest-neighbor hopping parameters (Figs. 3 and 4) are reported for transport context but do not affect the non-additivity conclusions.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1–3]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a total-energy cross term evaluated at the *identical* exchange-correlation functional level under joint biaxial (ϵ, δ) load. Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [4]. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [8], so external biaxial load couples to an already distorted cage baseline. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 9–11], while related 2D fullerene networks show anisotropic mechanical and tribological response under load [12, 13] and endohedral qHP C₆₀ cage filling reshapes band gaps [14]; none of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load, motivating explicit coupled grids rather than an additive superposition of composition and strain effects. Non-additive coupling is documented in other two-dimensional networks [1]; a quantitative *total-energy* synergy order parameter $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a single PBE+D3 protocol across fifteen periodic (n, δ) data points at $+3\%$ strain.

We address three questions on this model network: (1) how opposite linear strain coefficients α (meV per 1%

biaxial strain of the *total* cell energy) emerge for donor vs. acceptor dopants; (2) how large the omitted synergy cross term $|\mathcal{S}|$ (meV/atom) is when strain and substitution are applied jointly; and (3) whether $|\mathcal{S}|$ tracks $|\alpha|$ or instead reflects local structural distortion. Using $\gtrsim 90$ converged CP2K tasks (Table I), we define $\mathcal{S}$ on matched tetramer and periodic PBE+D3 grids (fifteen (n, δ) points at $+3\%$ strain) and report opposite donor/acceptor α on the tetramer strain grid, demonstrating that the largest $|\mathcal{S}|$ does not coincide with the smallest tetramer $|\alpha|$ for size-mismatched P. *Local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what linear α alone would suggest, so *relative* stability margins from $\epsilon = 0$ substitution energies or strain-only elastic increments need not survive a coupled load even if they do not reorder dopants with widely separated E_{sub} values. Marcus λ and hopping observables (Figs. 3 and 4) characterize transport-relevant quantities but are not used to infer $\mathcal{S}$.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All calculations use CP2K [15] with PBE [16] and Grimme DFT-D3 (Becke–Johnson damping) [17]; Table I summarizes basis sets, SCF convergence criteria, and the full task inventory. We do not employ the Koopmans-compliant rVV10 approach used in Ref. 4, and instead adopt a uniform PBE+D3 protocol suitable for large-scale parameter scanning, at the cost of absolute accuracy in polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is denoted by the calligraphic symbol $\mathcal{S}$, with units of meV per atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

Rigid-strain protocol. Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are *upper bounds* at fixed coordinates, not equilibrium thermodynamic quantities. Table II benchmarks four fixed-cell geometry-optimization corners for tetramer $P@+3\%$ to test whether $\mathcal{S}$ retains *sign* after ionic relaxation *within the same tetramer grid* (magnitude ratios are not quantitative until a matched-functional rigid reference exists); it does *not* validate periodic $n=4$ magnitudes, which differ in dopant concentration and boundary conditions. Main-text $|\mathcal{S}|$ values therefore bound—rather than predict—meV/atom-scale non-additivity until Table II converges.

Plane-wave cutoff (size scaling). Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table I); comparisons at fixed n always share the same cutoff. Cross- n trends therefore mix a modest plane-wave cutoff change at $n = 6$; we do *not* interpret the reported N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ as a physical size effect until the matched 400 Ry $n=6$ control in Table I completes.

C. Substitutional doping and grids

Tetramer models ($4 \times C_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates); these dopant configurations are not thermodynamically optimized ground states. A second tetramer placement ensemble (18 single-point energy tasks; Table III) tests whether tetramer α and $\mathcal{S}$ are placement-sensitive at $\sim 5\%$ nominal doping. All periodic supercells propagate the same dopant site map (one substituent per C_{60}); quantitative $|\mathcal{S}|$ on the fifteen-point grid is therefore configuration-specific, whereas donor/acceptor sign trends for α are discussed as element-intrinsic qualitative expectations pending Table III. Periodic scaling cells place one substituent per C_{60} (n dopants among $60n$ lattice atoms, ~ 1.67 at.% for all n), contrasting with the tetramer grid ($\sim 5\%$ nominal). Tetramer α , gaps, and tetramer IPR/ J trends (Fig. 4) are therefore not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ converged neutral tasks support the protocol-audited claims below (Table I); rigid-strain $|\mathcal{S}|$ magnitudes are upper bounds until Tables II and III complete. Charged-state geometry optimizations (UKS, multiplicity 2, EPS-SCF = $10^{-6} E_h$) supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^{\pm}$). Eight

vertical single-point energy evaluations compute $E_{\text{vert}}^{\pm}$ at the *neutral*-geometry coordinates with charged UKS settings matched to the relaxed state (EPS-SCF = $10^{-7} E_h$ by default; B $q = -1$: $10^{-6} E_h$), giving $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ following Ref. 4. These Marcus diagnostics appear only in Fig. 3 and do not enter Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table IV). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is the total-energy difference per dopant site at fixed strain *without* elemental chemical-potential corrections; it is *not* a thermodynamic standard formation enthalpy ($\sim$ eV-scale tabulated values) and is used only for relative ranking within this protocol. In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows under the rigid upper-bound protocol, the magnitude of this additive omission is best gauged against the substitution-energy scale: at $n = 4$ and $+3\%$ strain, P has $|\mathcal{S}| = 23.7$ meV/atom while the per-atom substitution scale is $|E_{\text{sub}}| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_{\text{sub}}| \approx 16$ eV per P site, $n_{\text{dop}}=4$ substituents in the $n=4$ cell, Table IV)—i.e. $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}} \sim 3\%$ as an *order-of-magnitude* ratio only, because tetramer and periodic dopant concentrations differ; this ratio is not used as a core nonlinearity test.

Non-additive coupling metric. We quantify strain-doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

When $\alpha_{\delta} = dE/d\epsilon$ denotes the *total*-cell linear strain coefficient (meV per 1% strain) at fixed n , a first-order estimate of the per-atom cross term is $\mathcal{S}(\epsilon, \delta) \approx (\alpha_{\delta} - \alpha_0) \epsilon / N_{\text{atoms}}$ at small ϵ under the rigid-strain protocol; tetramer and periodic n use different N_{atoms} and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

Periodic-cell $\mathcal{S}$ values are extracted from converged CP2K total energies (Table I); the fifteen-point $+3\%$

strain audit accompanies the open data release. We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Fig. 4). Following Capobianco *et al.* [4], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR is $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75\text{--}81$ meV reported for pristine qHP networks [4], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding. Marcus reorganization energies follow the vertical–adiabatic decomposition [4]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Fig. 3. Charged geometry optimizations supply adiabatic IP/EA; vertical single-points at the neutral geometry yield Marcus $\lambda^\pm$ (Fig. 3).

1. DFT calculator settings

All total-energy and electronic-structure calculations use CP2K [15] with PBE [16] and Grimme DFT-D3 (Becke–Johnson damping) [17]. Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). The plane-wave cutoff is 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table I); the atom-centered basis (DZVP-MOLOPT-SR-GTH) is unchanged across n . Tetramer models ($4 \times \text{C}_{60}$, 240 atoms) use B/N/P substitution at nominal 5% (12–13 dopant sites from archived coordinates). Periodic supercells place one substituent per C_{60} unit ($n = 1\text{--}8$), giving n dopants among $60n$ lattice atoms (~ 1.67 at.% for all n).

2. Marcus vertical step and post-processing

Charged-state geometry optimizations supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^\pm$). Eight vertical single-point energy evaluations compute $E_{\text{vert}}^\pm$ at the *neutral*-geometry coordinates with the charged UKS settings of the relaxed state, following Ref. 4; $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ (Fig. 3). π -projected densities of states use a 0.05 eV Gaussian broadening aligned to the Fermi level (Fig. 2). Inverse participation ratios and hopping integrals J are defined in Sec. II and Fig. 4; numerical values are listed in the Data Availability repository.

E. Validation benchmarks

The full calculation map (converged tasks, main-text usage, and pending controls) is in Table I; ionic-relaxation energies (Table II) and the alternate-placement grid (Table III) appear there when complete. Three validation benchmarks—rigid vs. relaxed tetramer $\mathcal{S}_P$ (Table II), alternate dopant placement (Table III), and a matched 400 Ry $n=6$ N control (Table I)—gate interpretation of the N $\mathcal{S}(n)$ sign change cited in Secs. III C and IV. Table II has completed the pristine fixed-cell geometry-optimization corners at $\epsilon=0$ and $+3\%$ ($E = -1363.9551$ and $-1363.9550 E_h$); $\mathcal{S}_{\text{relaxed}}$ and sign retention await the P-doped corners ($P@ \epsilon=0$ running; $P@ +3\%$ queued).

Table IV and Sec. III B report E_{sub} using the substitution-energy definition above (no chemical-potential reference; not a formation enthalpy).

III. RESULTS

We organize the results around total-energy non-additivity under the rigid-strain protocol (Sec. II B): Sec. III A summarizes band-edge context; Sec. III B reports linear strain coefficients, E_{sub} , and geometric strain response (Table V); Sec. III C gives $\mathcal{S}(n)$ at $+3\%$ strain across dopants and supercell sizes (Table I); Sec. III D reports Marcus λ and IPR/ J after the $\mathcal{S}$ audit.

A. Electronic structure under strain and doping

Figure 1(a,b) summarizes tetramer HOMO–LUMO gaps (archived tetramer electronic-structure set) and representative π -DOS (B @ $\epsilon=0$; color scheme follows Ref. 4). At $\epsilon=0$, B is near a closed π gap ($E_g \approx 0.03$ eV), N doping pins the Fermi level inside the conduction band ($E_g \approx -0.14$ eV), and P remains gapped ($E_g \approx 0.05$ eV); tensile strain narrows the pristine gap (~ 23 meV/% on the three available strain points in the dataset). Donor vs. acceptor ordering of the $\epsilon=0$ gaps tracks the opposite α signs in Sec. III B, but band-edge slopes are not used to infer $\mathcal{S}$.

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). On the tetramer grid (Table IV), substitution opens an order-of-magnitude spread in α : N (-303.2 meV/%, tension stabilizes the doped cell), B ($+56.9$ meV/%, tension destabilizes), and P (-19.1 meV/%, weakly stabilizing). The opposite sign for N vs. B persists across the $\epsilon \in \{-5, \dots, +5\}\%$ grid at this fixed placement and is not an artifact of a single strain point; Table III will test whether the slopes are placement-sensitive. At $\epsilon = +5\%$, N remains

TABLE I. Summary of converged DFT calculations and correspondence to main-text results. The ~ 90 tasks cited in the main text are the converged calculations that directly support quantitative claims; the full archive includes additional preliminary grids (rows marked “not used”).

Calculation set	Model	Converged tasks	Main-text usage	Notes
Early strain probe	$2 \times C_{60}$	5	not used	superseded by tetramer grids
Single-molecule doping	C_{60}	11	not used	superseded
Early electronics	$2 \times C_{60}$	24	not used	superseded by tetramer π -DOS
Polaron IPR/ J tetramer dimer	tetramer	4	Fig. 4	pristine/B $\times \epsilon=0, +3\%$
Tetramer strain energies	tetramer	24	Table IV, α	rigid strain
Optimal-design grid	tetramer	20	not used	archive only
Tetramer π -DOS/gaps	tetramer	12	Fig. 2; main panels	
Geometry references	tetramer relax	6	charged-state inputs	reference geometries
Charged polarons	tetramer $q = \pm 1$	12 geometry opt.	Fig. 3	Marcus $\lambda^\pm$ and adiabatic IP/EA
Size scaling	$n \times C_{60}$	40 (+1 pending)	main $\mathcal{S}(n)$	400 Ry ($n \leq 4$), 350 Ry ($n \geq 6$); cutoff400 contr
$n=6$ cutoff control	$6 \times C_{60}$ N @+3%	pending	N sign benchmark	400 Ry matched control (pending)
Ionic relaxation (tetramer)	tetramer	2/4 geometry opt.	Table II	fixed-cell; pristine $\epsilon=0, +3\%$ converged

TABLE II. Ionic-relaxation benchmark for tetramer $P@ + 3\%$ (four fixed-cell geometry-optimization corners, same biaxial cell). This benchmark is *sign-qualitative only*: the rigid reference uses legacy PBE without D3, whereas relaxed corners use PBE+D3, so $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$ is *not* a quantitative retention ratio until a matched-functional rigid grid is recomputed. We test whether $\mathcal{S}$ retains *sign within the tetramer grid only*; it does *not* calibrate periodic main-text $\mathcal{S}(n)$, which uses different dopant concentration and boundary conditions. Rigid reference energies: archived fixed-coordinate tetramer single-point grid (legacy PBE, no D3). Relaxed energies: PBE+D3 fixed-cell geometry optimization (400 Ry) from archived inputs in the open repository. $\mathcal{S}$ follows Eq. (1) with $N=240$ atoms.

Quantity	Rigid (legacy tetramer)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	−1363.5512	−1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	−1363.5522	−1363.9550
$E(P, \epsilon=0) (E_h)$	−1370.4965	pending
$E(P, \epsilon=3\%) (E_h)$	−1370.4890	pending
$\mathcal{S}$ (meV/atom)	+0.96	pending
Sign of $\mathcal{S}$ vs. rigid	pending (2/4 geometry opt.; $P@ \epsilon=0$ running)	

the most tension-stabilized dopant (α most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative α , foreshadowing why its linear coefficient poorly predicts $|\mathcal{S}| = 23.7$ meV/atom for P at $n=4$ under +3% strain (Sec. III C). Substitution energies at $\epsilon = 0$ span $E_{\text{sub}} = +73.4$ eV/dopant (B), -115.3 eV/dopant (N), and -15.8 eV/dopant (P)—raw total-energy differences without chemical-potential reference, not standard formation enthalpies—with $n_{\text{dop}} = 12\text{--}13$ counted from input coordinates (Table IV). Figure 1(c) highlights the α split; ranking dopants solely by $|E_{\text{sub}}|$ at $\epsilon=0$ is misleading once +3% strain is applied. Table V links this mechanical spread to local geometry: P shows negligible strain-induced change in mean nearest C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at +3% load), foreshadowing the α – $\mathcal{S}$ decoupling quantified in Sec. III C (Discussion mechanistic synthesis).

C. Non-additive strain–doping coupling

Figure 1(d) tracks protocol $\mathcal{S}(n)$ (rigid-strain upper bounds) with one dopant per C_{60} at +3% biaxial strain.

At $n = 4$ under the fixed periodic dopant map (alternate placements not yet computed), $\mathcal{S} = -6.2$ meV/atom (B), $+3.0$ meV/atom (N), and -23.7 meV/atom (P); P shows the largest magnitude despite the smallest $|\alpha|$ on the tetramer grid (Table IV). Across fifteen periodic points ($n \in \{1, 2, 4, 6, 8\}$, B/N/P), $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom at larger n for B and P, whereas N is smaller in magnitude at $n = 4$ but comparable at $n = 1$. For N, $\mathcal{S}$ changes sign between $n = 6$ ($+2.4$ meV/atom) and $n = 8$ (-2.3 meV/atom) at the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until the matched 400 Ry $n=6$ control in Table I and Tables II and III complete (Sec. II E). Core size-scaling claims focus on $n \leq 4$ at matched 400 Ry cutoffs; B and P show monotonic $|\mathcal{S}|$ decrease from $n = 1$ toward $n = 4$, whereas $n \geq 6$ trends are reported for completeness only and mix the 350 Ry production cutoff. We report provisional $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$ fits in the open data release but refrain from drawing conclusions about the asymptotic $\mathcal{S}_\infty$ value until cutoffs are matched across n . All periodic $\mathcal{S}$ values quoted in this subsection use the fixed dopant-site map and are tabulated in Table I (fifteen $n \times$ dopant points at +3% strain). Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *sta*

TABLE III. Alternate dopant-placement sensitivity (*tetramer only*, rigid strain, PBE+D3). The *reference* placement defines all main-text tetramer α and the periodic supercell site map (one substituent per C_{60}). An *alternate* independent placement repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 single-point tasks) to test whether tetramer α and $\mathcal{S}$ at +3% are placement-sensitive at $\sim 5\%$ nominal doping. Periodic $\mathcal{S}(n)$ is *not* tabulated here because concentration and boundary conditions differ; main-text periodic magnitudes are configuration-specific until alternate periodic placements are computed.

Dopant	Placement	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)	Status
B	reference	+56.9	$\sim +0.7$	verified (Table S1)
N	reference	-303.2	~ -3.8	verified
P	reference	-19.1	~ -0.2	verified
B	alternate	pending	pending	pending
N	alternate	pending	pending	pending
P	alternate	pending	pending	pending

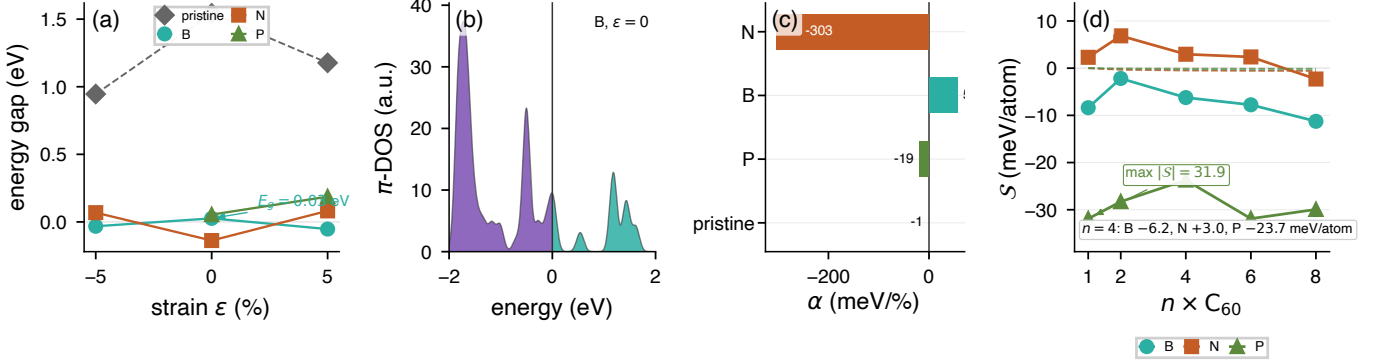


FIG. 1. Coupled (ϵ, δ) response in doped qHP C_{60} (PBE+D3). (a) HOMO-LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon=0$ (color scheme follows Ref. 4). (c) Mechanical α (Table IV). (d) $\mathcal{S}(n)$ at +3% (max $|\mathcal{S}|=31.9$ meV/atom for P @ $n=1$; at $n=4$: -6.2, +3.0, -23.7 meV/atom for B/N/P); Marcus λ and IPR/ J in Figs. 3 and 4.

bilizes the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n=4$) means sequential screens *overestimate* stability under load.

D. Polaron transport diagnostics

Figure 3 reports Marcus reorganization energies from twelve charged-state geometry optimizations and eight vertical single-points at the neutral geometry (Sec. IID 2).

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR $75.0 \rightarrow 24.6$) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

IV. DISCUSSION

Context. Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–14, 18, 19], the distinction here is a periodic *total-energy* audit of joint biaxial (ϵ, δ) load

[Eq. (1)] on a rigid-strain upper-bound protocol at the identical PBE+D3 functional level. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [19]; isomer-engineered C_{70} network designs similarly tune band edges through topology choice rather than joint substitutional–mechanical load [20]; we instead quantify the omitted additive cross term $\mathcal{S}$ in total energy. Quasi-harmonic thermal-expansion studies of qHP monolayers report anisotropic positive and negative lattice response tied to interfullerene bond topology [10, 18], whereas our audit holds fractional coordinates fixed at zero-temperature total energy (upper bound pending Table II). Strain-engineering studies on smaller fullerene graphullerenes instead emphasize magnetic or half-metallic transitions [21], which is orthogonal to the neutral total-energy audit pursued here. Band-edge, spin-resolved, and hopping metrics (Fig. 4) contextualize mechanism but are not used to infer $\mathcal{S}$.

Mechanistic synthesis. (i) **Opposite α for N vs. B.** N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^N < E_g^B < E_g^P$ within PBE; Sec. III A) and dopant-resolved π -DOS in Fig. 1(b) and

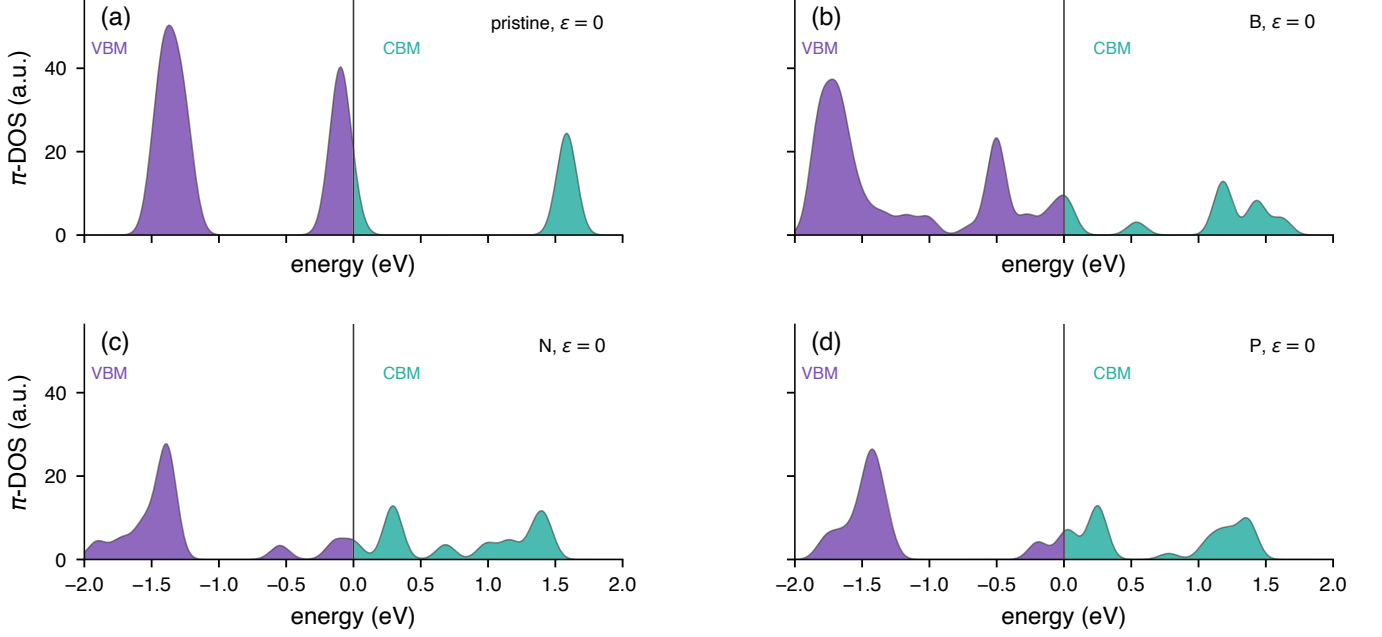


FIG. 2. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [4]).

TABLE IV. Total energies, linear strain sensitivities, and substitution energies for tetramer models (PBE+D3, rigid-strain protocol). $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ with no chemical-potential reference (not a standard formation enthalpy). n_{dopants} is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12).

System	n_{dop}	$E(\epsilon=0)$ (E_h)	$E(\epsilon=+5\%)$ (E_h)	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

Fig. 2 align with donor vs. acceptor π -counting, while total-energy α spans ~ 360 meV/% between N and B on the tetramer grid (Table IV). The opposite N vs. B sign is reported for this fixed placement across $\epsilon \in \{-5, \dots, +5\}\%$ (Table IV); Table III repeats the full tetramer strain grid for an alternate placement to test slope sensitivity at $\sim 5\%$ nominal doping without mixing periodic boundary conditions. Mayer bond order and Bader charge partitioning along the strain path are not yet computed (Limitations); periodic $\mathcal{S}(n)$ magnitudes are likewise not averaged over alternate supercell placements until such audits exist. **(ii) Large $|\mathcal{S}|$ for P despite small $|\alpha|$.** P carries the largest covalent-radius excess (+30 pm vs. C; Table V) and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B/N) and the smallest site-to-site spread at +3% ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table V). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than to a uniform elastic response, yielding a cross term invisible in separate scans but dominant when both strain

and substitution are applied simultaneously—consistent with P’s intermediate E_{sub} yet the largest $|\mathcal{S}|$ at $n=4$ (23.7 meV/atom; Table I) despite weak tetramer $|\alpha|$. Sec. III B presents this Table V geometric decoupling before the $n=4$ $\mathcal{S}$ ranking in Sec. III C: tetramer $|\alpha|$ ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the largest cross term. **(iii) N $\mathcal{S}$ sign change with n .** With one substituent per C_{60} , electrostatic fields dilute at large n ; any N sign flip between $n=6$ and $n=8$ is confounded with the 400→350 Ry cutoff step and remains uninterpreted (Sec. II E). Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n). **(iv) Geometric vs. electronic nonlinearity.** For P, Table V shows that rigid tensile load barely moves the mean dopant–C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å) yet $\mathcal{S}$ remains the largest at $n=4$ (23.7 meV/atom), whereas B/N exhibit $\sim 10^2 \times$ larger $\Delta \bar{d}$ at +3% but smaller $|\mathcal{S}|$ (6.2 and 3.0 meV/atom). At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer

TABLE V. Local dopant–cage structure from archived tetramer XYZ (mean nearest C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm). Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ summarizes how tensile load couples to pre-existing distortion (Discussion mechanistic synthesis (ii),(vi)). Values are computed from archived tetramer XYZ coordinates in the open data release; Mayer bond order and Bader charges are not included.

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

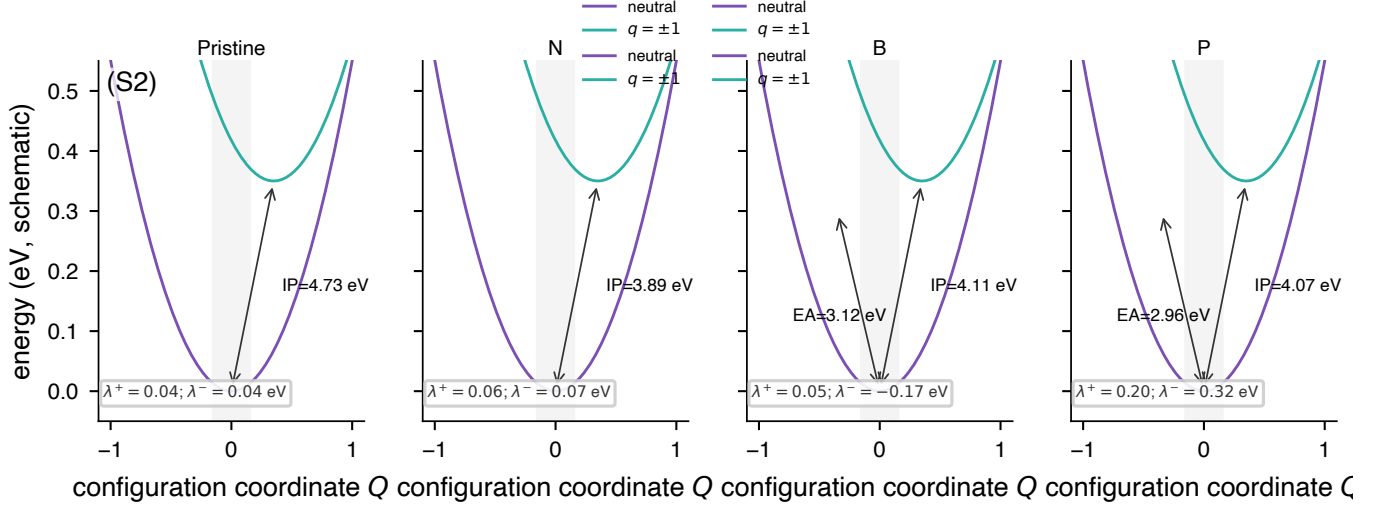


FIG. 3. Marcus reorganization energies (12/12 charged-state geometry optimizations; eight vertical single-points at the neutral geometry). Pristine $\lambda^\pm=0.041/0.042$ eV; N $\lambda^+=0.058/\lambda^-=0.065$ eV; B $\lambda^+=0.051$ eV; P $\lambda^+=0.200/\lambda^-=0.315$ eV. The electron-channel reorganization energy λ^- for B doping is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

α by two orders of magnitude ($|\mathcal{S}| \gg \alpha \times 3\%/N_{\text{atoms}}$; Methods energy observables), even though $|\alpha|$ ranks P as the weakest strain responder on the tetramer grid. **(v) Argument scope (concentration and boundary conditions).** We do *not* validate $\mathcal{S}$ by extrapolating tetramer α to periodic $n=4$: the $\sim 5\%$ tetramer vs. ~ 1.67 at.% periodic concentrations and tetramer vs. periodic boundary conditions are unmatched, so a first-order $(\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ estimate is excluded from the argument. Non-additivity is assessed only through Eq. (1) on matched four-corner grids at common (ϵ, δ) ; P in particular requires explicit coupled evaluation rather than sequential α screening. **(vi) Decomposition outlook.** The Table V decoupling indicates that the cross term is not reducible to uniform bond stretching alone; electronic rehybridization and pre-stored cage strain likely contribute, but their relative weights are not partitioned here without Mayer/Bader analysis along the strain path (Limitations).

(vii) Significance versus DFT uncertainty. $\mathcal{S}$ is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. At $n=4$ and +3% strain, $|\mathcal{S}| = 23.7$ meV/atom for

P is $\sim 3\%$ of the per-atom E_{sub} scale (Table IV) but still $\gg$ rigid per-atom strain increments from α . Large absolute uncertainties in PBE E_{sub} therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

Design implications. At $n=4$ and +3% strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates differ by approximately a factor of four in the load-dependent *margin* even though widely separated E_{sub} ranks at $\epsilon=0$ (e.g., B vs. P; Table IV) are unchanged. Dopants with larger covalent-radius excess (P: +30 pm vs. B: +7 pm; Table V) show the largest protocol $|\mathcal{S}|$ under fixed coordinates; *size-mismatched* substituents may warrant coupled (ϵ, δ) evaluation rather than one-parameter pipelines once Table II confirms retention after relaxation. The fifteen-point periodic grid tabulates protocol $|\mathcal{S}|$ at each (n, δ) as an audit of additive omission under the rigid protocol (Table I), not as quantitative design targets. At +3% strain, P carries the largest additive-approximation error in protocol $\mathcal{S}$ ($|\mathcal{S}| = 23.7\text{--}31.9$ meV/atom from $n=4$ to $n=1$ under fixed coordinates); qualitative co-design should treat P under tension as a *coupled* stability problem pending Table II, not $|E_{\text{sub}}|$ at $\epsilon = 0$ alone. Where

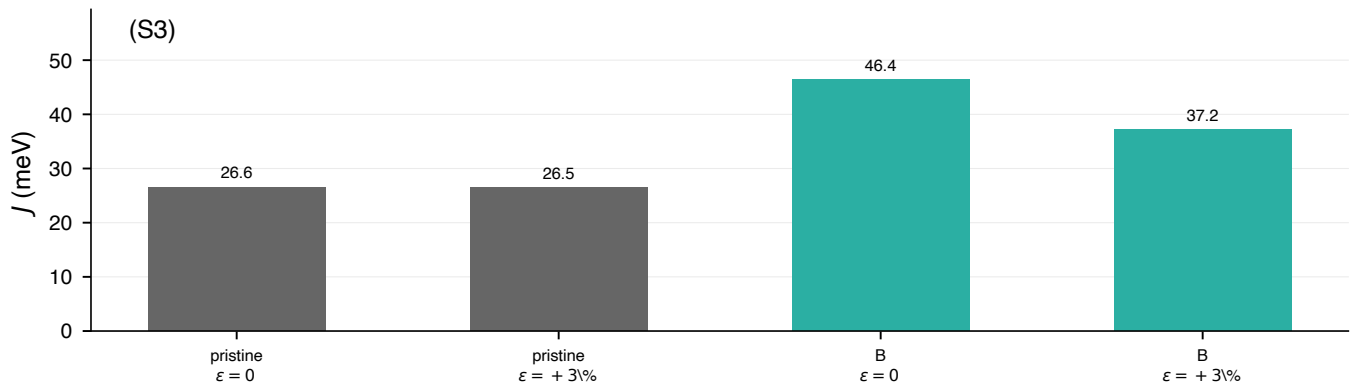


FIG. 4. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3).

$S < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $S > 0$ (N at $n=4$), sequential screens overestimate stability under load. For chemistries with comparable $|E_{\text{sub}}|$ or in multi-dopant competition, omitting S can shift margins by several percent of the per-atom substitution scale—much larger than the rigid strain increment from α —even when the global dopant rank at $\epsilon=0$ is unchanged. Gap closing [Fig. 1(a)] does not track $|S|$; we do not compute S_{gap} , so photonic or transport targets cannot be inferred from total-energy S alone. For B at $n = 4$, both negative S and positive α imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating S . For N at $n=4$, negative tetramer α and positive S present a second sign conflict: tension stabilizes the doped tetramer in isolation, yet the periodic coupled audit reports a positive cross term that sequential additive models fail to capture at ~ 1.67 at.% doping. N S vs. n is reported for completeness but excluded from design rules until Tables I–III controls converge.

Validation protocol (Tables I–III). Three validation benchmarks characterize protocol limits without altering the periodic magnitudes quoted above. **Table II** runs fixed-cell geometry optimization on the four Eq. (1) corners for tetramer $P@ +3\%$ only; because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3, the acceptance criterion is *sign retention* of S after ionic relaxation within that tetramer grid—not a quantitative $|S_{\text{relaxed}}|/|S_{\text{rigid}}|$ ratio until a matched-functional rigid grid is recomputed. It does *not* rescale periodic $S(n)$ because concentration and boundary conditions differ. **Table III** tests tetramer α and tetramer $S_{+3\%}$ for alternate vs. reference placement; qualitative claims about opposite N/B α signs are configuration-specific until this grid converges. **Table I** (cutoff400 row) isolates the 400→350 Ry confound for $6 \times C_{60}$ N @ $+3\%$ before any $n \geq 6$ design statement is restored.

Polaron transport. The Marcus workflow in Figs. 3

and 4 parallels Ref. 4—charged-state geometry optimizations ($E_{\text{opt}}^{\pm}$) plus vertical single-points at the neutral geometry ($E_{\text{vert}}^{\pm}$)—at the uniform PBE+D3 level used for S . Recent work shows that charged-supercell and neutral-defect semilocal routes to polaron self-interaction correction are formally equivalent when the polaronic structure is held fixed [22]; our Marcus λ protocol instead follows the charged vertical/adiabatic split of Ref. 4 without piecewise-linear or γ/μ_{DFT} tuning, and remains orthogonal to the S audit. Where that study linked hopping J and frontier-orbital localization to mobilities in pristine qHP networks, we retain the same observables for comparability (Fig. 4) but do *not* extend their mobility-centric ranking to neutral total-energy margins. Endohedral band-gap tuning [14] and substitutional S address a second orthogonal knob (cage filling vs. cage substitution) and are not interchangeable for sequential-screen corrections.

Limitations and outlook. Three protocol limits are actively benchmarked (Table I; Sec. IIE): rigid strain (Table II), tetramer alternate-placement sensitivity (Table III), and mixed 400/350 Ry cutoffs across n (Table I cutoff400 control). All periodic $S(n)$ use a single fixed dopant configuration; magnitudes are not averaged over placement ensembles, and periodic alternate-site audits are not yet computed. Additional limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10; S_{gap} not quantified; Mayer bond-order and Bader charge partitioning along the strain path are not computed (Table V reports geometric metrics only); the electron-channel reorganization energy λ^- for B doping is unphysical under the fixed neutral-geometry vertical protocol and is excluded from rate estimates (as further discussed in the Limitations subsection); mobility tensors are not computed—polaron transport anisotropy in related fullerene networks is treated only comparatively [4, 6] (Figs. 3 and 4).

V. CONCLUSIONS

Sequential strain-doping workflows in covalent molecular networks can omit a meV/atom-scale total-energy cross term when local dopant distortion couples nonlinearly to external biaxial load. In qHP C₆₀ graphullerene we answered the three Introduction questions under a uniform PBE+D3 rigid-strain protocol: **(1)** opposite donor/acceptor α (N vs. B; Table IV); **(2)** meV/atom-scale omitted cross terms on a fifteen-point periodic grid (upper bounds at fixed coordinates; Table I); **(3)** $|\mathcal{S}|$ decoupled from tetramer $|\alpha|$ for size-mismatched P via pre-existing cage distortion (Table V). Ionic-relaxation, alternate-placement, and matched-cutoff controls (Tables I–III) will test whether these audits remain relevant after relaxation and placement averaging. Coupled (ϵ, δ) screening is advisable for size-mismatched substituents and close substitution-energy competitors, complementing strain-only mechanical [12], derivative-network [11], and endohedral band-edge [14] axes. Future work extends alternate-placement validation to periodic cells, adds Mayer/Bader strain-path partitioning, and explores

polaron-aware semilocal corrections beyond the Marcus vertical protocol.

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DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including converged computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text. Pending ionic-relaxation, alternate-placement, and matched-cutoff validation tasks are documented in Tables I–III. A tagged release matching the manuscript revision will be issued upon acceptance.

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